Data Preparation Summary: Frequency and Severity

IAA Hackathon · as of Oct 18, 2026 · original data, analysis, adjustment and open issues by peril

Frequency = completeness-adjusted claims per 1,000 exposure. Severity = mean amount gross of excess. Figures are 2021 → 2024.

0. Common cleaning (affects frequency and severity)

Step	Original	Analysis	Adjustment
Claim–policy matching	Claims carry only policy_id	Match by loss date within the period; consecutive periods are exactly 1 day apart	Match rate ≥ 99.7%; claims outside any period excluded
Exposure	0.5% on a 360-day basis, 0.1% recorded in days	Compared with days / 365 from period dates	Replaced with date-based value; total exposure −25 to −28% (frequency denominator)
Duplicate claims	604 records differ only in claim_id; 600 in 2022	All fields compared; no ambiguous near-duplicates	Removed; 2022 frequency and cost were ~2% too high
Repudiated claims	All amounts zero	Only closed, open and repudiated statuses exist	Excluded; frequency counts claims with net cost > 0

1. Motor

Frequency

Peril	Original data	Analysis	Adjustment	Open issue
bodily_injury	11.4 → 10.5. Median report lag 112 days; 2024 is 56% complete	Unreported claims estimated from 2021 lags; adjusted frequency is nearly flat	Exposure weighted by completeness	Whether 2021 lags still apply in 2024
own_damage	63.7 → 59.8; one-off drop in 2023	Mix change tested; the drop appears within every group	Pre-2023 × 0.93 (level shift)	Cause of the drop unknown
third_party_prop	43.5 → 40.8; lower from 2023	Same test	Pre-2023 × 0.95	Same
glass_hail	19.0 → 20.7; +3% a year	Not analysed	None	Weather (hail_days) link not checked
theft	7.9 → 7.7; flat	—	None	—

Severity

Peril	Original data	Analysis	Adjustment	Open issue
bodily_injury	21.2k → 17.7k; one 11.9m claim	Quarterly and within-365-day comparisons: step at 2023 Q1 (−19%), BI only. 'Late claims larger' and 'reserves too low' both rejected	Pre-2023 × 0.83; post-reform trend 0%; cap 500k + loading	Post-reform trend (1 year observed); case-reserve development (2024 ~2% low)
third_party_prop	3.3k → 4.1k (+8% a year)	Trend measured gross of excess	+8% a year	Whether excess applies to liability (~4.6% of motor)

own_damage	2.3k → 2.5k (+4% a year)	Net-of-excess median +13% a year: deductible leverage effect	+4% a year, applied gross of excess	Expected recoveries on open claims not reflected
glass_hail	945 → 1,191 (+8% a year)	Same	+8% a year	—
theft	6.3k → 6.8k (+3% a year)	Same	+3% a year	Pattern 6.3k, 6.2k → 6.9k, 6.8k looks like a 2023 step

2. Buildings

Frequency

Peril	Original data	Analysis	Adjustment	Open issue
subsidence	2.00 → 2.82 (+12% a year). Median lag 85 days; 2024 is 69% complete	Top dry_summer_index quintile has 3.1x the frequency of the bottom; drier summers in 2023–24	No trend; weather features	2024 completeness estimate (largest buildings uncertainty)
flood	3.94 → 3.66	Top rainfall quintile 2.5x; falls with rainfall	Weather features	—
storm	12.6 → 13.1 (14.2 in 2023)	Top wind quintile 1.9x, but yearly moves opposite to average wind	Weather features	2023 rise not explained by weather
fire	1.68 → 2.10 (+8% a year)	Top wildfire quintile 1.8x; yearly rise not explained by weather	None	Whether to add a frequency trend (by validation)
escape_of_water	17.6 → 17.9; flat	—	None	—
accidental_damage	6.2 → 6.8 (+3% a year)	—	None	Small rise, not checked

Severity

Peril	Original data	Analysis	Adjustment	Open issue
escape_of_water	5.1k → 6.5k (+8% a year)	Quarterly medians rise gradually: inflation-type trend	+8% a year	—
flood	28.5k → 34.9k (+7% a year)	Noisy by quarter but rising	+7% a year	Trend may include event size (weather)
storm	3.1k → 3.9k (+8% a year)	Flat until mid-2023, then rising	+8% a year	Recent rise; 8% may be high (sensitivity)
subsidence	22.1k → 27.1k (+7% a year)	Mean only	+7% a year	Quarterly pattern not checked
fire	54.3k → 50.9k (−2%)	Driven by large claims; top 5 claims are all fire	Trend 0%; cap 300k + loading	Cap level; link to sum_insured
accidental_damage	1.2k → 1.3k (+3% a year)	—	+3% a year	—

3. Contents

Frequency

Peril	Original data	Analysis	Adjustment	Open issue
theft	12.4 → 13.5 (+3% a year), rising every year	Not a weather peril	None	Whether to add a frequency trend (by validation)
flood	2.87 → 2.53; volatile	Explains low 2022 cost	Weather features	Weather link not checked for contents
escape_of_water	9.0 → 9.6	—	None	—
accidental_damage	15.4 → 15.8	—	None	—

Severity

Peril	Original data	Analysis	Adjustment	Open issue
theft	3.5k → 4.0k (+4% a year)	Measured on the mean	+4% a year	Quarterly pattern not checked
flood	9.5k → 10.9k (+5% a year)	Large rise in 2023	+5% a year	Step vs trend not checked
escape_of_water	2.6k → 2.8k (+3% a year)	—	+3% a year	—
accidental_damage	763 → 823 (+3% a year)	—	+3% a year	—

Development: all perils ≥ 95% complete in 2024. Cap: none (largest claim 111,300).

4. Open issues (by priority)

#	Issue	Impact	Affects	How to resolve
1	Excess on liability perils	~4.6% of motor	Severity	Kaggle forum answer or leaderboard comparison
2	Recoveries on open claims	Net cost of open own_damage and theft claims too high	Severity	Decide whether to apply closed-claim recovery rates (own_damage ~22%) to open claims
3	Buildings 2024 subsidence completeness	~+4% on buildings 2024	Frequency	Report as sensitivity
4	Fire and contents theft frequency trend	May under-predict 2025 for these perils	Frequency	Out-of-time validation on 2024
5	Case-reserve development	Motor 2024 ~2% low	Severity	Single snapshot; record as limitation
6	Motor theft severity: step or trend	Small	Severity	Quarterly check
7	Contents weather link and quarterly severity	Small	Both	Same code as buildings
8	Cap levels (motor 500k, buildings 300k)	Affects allocation only; totals kept by loading	Severity	Choose by validation